

The Canonical Koopman Lift Is Too Large: Dense Rational Spectrum for a Frobenius Suspension

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Abstract

An exact periodic-orbit zeta does not ensure that the canonical unitary lift of the same flow has a locally finite spectral determinant. We audit the unchanged constant-roof $\log 2$ suspension of the square-map Frobenius $F(a) = a^2$ on the discrete geometric points of $\mathbb{P}_{\mathbb{F}_2}^1$. On the Hilbert direct sum of its closed-point circles, Koopman translations form a strongly continuous unitary group whose self-adjoint Stone generator is $A_K = \bigoplus_x (-i d/du)_{\text{per}}$ on the full periodic Sobolev domain. Every family of finite, strictly positive component weights gives a unitarily equivalent representation. We prove

$$\sigma_p(A_K) = \frac{2\pi}{\log 2} \mathbb{Q}, \quad \sigma(A_K) = \sigma_{\text{ess}}(A_K) = \mathbb{R},$$

and every point eigenvalue has countably infinite multiplicity. The eigenvectors nevertheless form a complete pure-point basis; irrational reals are continuous-spectrum accumulation points in the operator-theoretic set decomposition, not a continuous part of the vector spectral measures. The resolvent is noncompact, every interval of positive width has infinite spectral-projection rank, and the Gaussian and absolute-value heat operators are not trace class, even after removal of the zero modes. Thus the natural lift passes complete-definition and self-adjointness tests but fails the local-finiteness gate. The primitive-orbit Hasse–Weil product and Deligne’s finite-dimensional cohomological Frobenius determinant belong to different operator ledgers and are not spectral determinants of this generator.

摘要

一个精确的周期轨道 zeta 函数，并不保证同一流的典范西提升具有局部有限的谱行列式。本文审计一个完全冻结的对象：离散化的 $\mathbb{P}_{\mathbb{F}_2}^1$ 几何点上的算术 Frobenius，以 $\log 2$ 为常值屋顶构造悬挂流。闭点圆周的 Hilbert 直和上，Koopman 平移构成强连续酉群；其 Stone 自伴生成元为 $A_K = \bigoplus_x (-i d/du)_{\text{per}}$ ，定义域是完整的周期 Sobolev 直和定义域。任意严格正且有限的分量权重都与典范测度酉等价。本文证明点谱恰为 $(2\pi/\log 2)\mathbb{Q}$ ，每个点特征值均具有可数无穷重数，且全谱与本质谱均为 $\mathbb{R}$ 。特征向量构成完备纯点基，但无理数仍是算子集合分解意义下的连续谱聚点；这不意味着向量谱测度含连续部分。由此，预解算子不紧，每个正宽区间的谱投影均为无穷秩， $e^{-tA_K^2}$ 与 $e^{-t|A_K|}$ 均非迹类；删除零模后结论仍不改变。因此，该自然提升通过完整算子定义与自伴性检验，却在局部有限性门槛处严格失败。轨道 Hasse–Weil 乘积及 Deligne 的有限维上同调 Frobenius 行列式属于不同算子账本，不能被改称为此 Koopman 生成元的谱行列式。

Keywords: Koopman group; Frobenius suspension; Stone generator; dense point spectrum; essential spectrum; compact resolvent; Hasse–Weil zeta.

1 Introduction

The canonical Hilbert-space lift of the frozen Frobenius suspension exists, is natural, and is self-adjoint. It is also too large for a locally finite spectral interpretation. This combination is useful: it converts an apparently promising Route-A unitary construction into an exact obstruction at the first Route-B spectral gate where local eigenvalue counting matters.

Koopman’s representation of a measure-preserving dynamics by composition operators is classical [3]. A measurable flow with invariant measure gives a one-parameter group of such operators, and strong continuity places the group under Stone’s theorem [6, 7]. The question here is not whether some self-adjoint operator can be written down. It is whether the operator forced by an already locked arithmetic flow has the spectral type required for the determinant aspirations that motivate arithmetic-flow research.

The underlying flow is the constant-roof suspension of the square-map Frobenius $F(a) = a^2$ on $\mathbb{P}^1/\mathbb{F}_2$, previously established as a finite-field positive control [9]. Its primitive orbit product is

$$\prod_{x \in |\mathbb{P}^1_{\mathbb{F}_2}|} (1 - e^{-s \deg(x) \log 2})^{-1} = \frac{1}{(1 - 2^{-s})(1 - 2^{1-s})}.$$

No roof, phase, potential, component coupling, or fitted coefficient is added in the present paper. The sole new step is the canonical Koopman lift on the same circles with the same clock.

The main result is a complete spectral classification. Every degree occurs among the closed points of $\mathbb{P}^1/\mathbb{F}_2$. A degree- d circle has Fourier frequencies $2\pi n/(d \log 2)$, so all denominators occur. More strongly, a rational frequency a/b is repeated on degrees kb at modes ka for every $k \geq 1$. The point spectrum is therefore dense and every point eigenspace is countably infinite-dimensional. The direct-sum spectrum and essential spectrum are both $\mathbb{R}$.

These facts settle the standard compactness questions without asymptotic estimates. The infinite zero eigenspace already prevents compact resolvent and trace-class heat. Removing it does not help, because any nonzero rational frequency has the same infinite multiplicity. Every interval of positive width contains such a frequency. A singleton irrational set, by contrast, has zero spectral projection; the interval qualifier is essential.

The result also resolves a ledger ambiguity. The Koopman generator acts on Fourier modes of L^2 observables. The orbit product counts primitive circles and repetitions. Deligne’s formula acts on finite-dimensional ℓ -adic cohomology [2]. Their common closed-point source does not provide an intertwiner, trace identity, or determinant identity. The same-object discipline developed in the companion trace audit forbids combining coordinates from these separate ledgers [10].

In the route vocabulary used by the research program, the paper proves `A4_UNITARY_OR_SCATTERING_CANDIDATE`, `B1_COMPLETE_OPERATOR_DEFINITION`, and `B2_SELF_ADJOINT`; it also proves `B3_FAIL`. The scoped result is `ROUTE_B_REJECTED` at Gate C. B4 and B5 are outside this limited audit and receive no verdict. No Hilbert–Pólya claim is made.

2 The frozen Frobenius suspension

Let

$$X = \mathbb{P}^1_{\mathbb{F}_2}, \quad S = X(\overline{\mathbb{F}_2})_{\text{disc}}, \quad F(a) = a^2, \quad \tau = \log 2.$$

The topology on S is explicitly discrete. It is a modeling choice inherited from the frozen classical candidate, not a topology of the scheme. We call $F(a) = a^2$ the square-map Frobenius convention throughout. Sources that place geometric and arithmetic Frobenius inversely may use the opposite

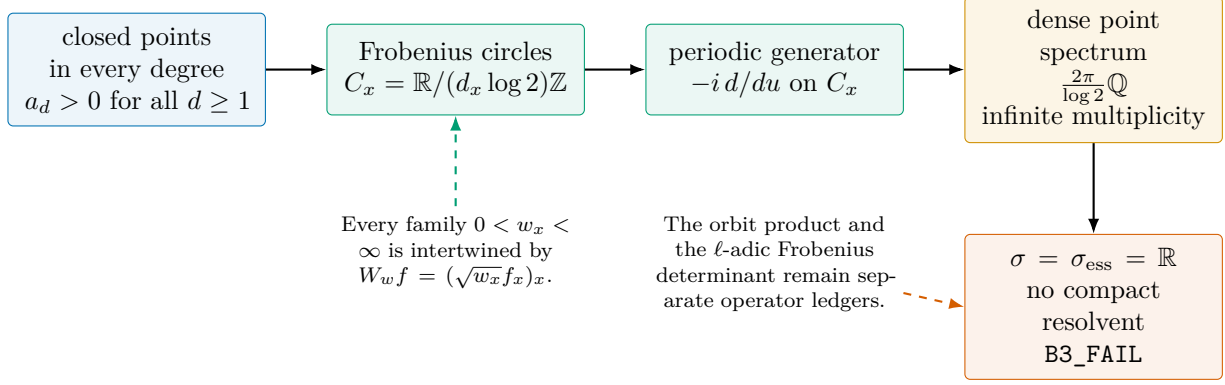


Figure 1: Exact implication chain for the frozen Koopman candidate. The positive-weight family is one unitary-equivalence class, while the orbit and cohomological ledgers do not become determinants of the Stone generator.

Table 1: Frozen definition of candidate FF-FROB-SUSP-P1-F2-KOOPMAN-P1.

Field	Canonical representative	Allowed equivalent family
Components	$C_x = \mathbb{R}/(d_x \log 2)\mathbb{Z}$	unchanged
Measure	$\mu_1 _{C_x} = du_x$	$\mu_w _{C_x} = w_x du_x, 0 < w_x < \infty$
Hilbert space	$\bigoplus_x^2 L^2(C_x, du_x)$	$\bigoplus_x^2 L^2(C_x, w_x du_x)$
Group	$(U_t f)_x(u) = f_x(u - t)$	same formula
Stone convention	$U_t = e^{-itA_K}$	same convention
Component action	$-i d/du$	same action
Boundary condition	periodic Sobolev trace	unchanged
Added data	none	positive constants only

name. Replacing F by F^{-1} only reverses each finite cycle and leaves its degree, suspension length, and every spectrum calculation below unchanged; the frozen return map itself remains $a \mapsto a^2$. The mapping torus and vertical flow are

$$M_F = (S \times \mathbb{R})/\mathbb{Z}, \quad n \cdot (a, u) = (F^n a, u - n\tau), \quad \phi^t[a, u] = [a, u + t]. \quad (1)$$

Closed points are the finite Frobenius orbits on geometric points [2, Sec. 1.4]. If x has residue degree d_x , its suspension component is the circle

$$C_x = \mathbb{R}/(L_x \mathbb{Z}), \quad L_x = d_x \log 2, \quad M_F \cong \coprod_{x \in |X|} C_x. \quad (2)$$

The flow on C_x is $u \mapsto u + t \pmod{L_x}$. Each component is open and closed, and the component set is countable.

The orbit identity quoted in the Introduction belongs to the classical ledger. It comes from the bijection between closed points and primitive circles and from the period identity $L_x = \deg(x) \log 2 = \log N(x)$. The present operator construction changes none of these data. Table 1 records every new analytic choice before the spectral calculation.

3 A closed point in every degree

The later spectral result needs only one component of each positive degree. For completeness, the exact degree count is recorded first. Let a_d be the number of degree- d closed points of X . The affine closed points are monic irreducible polynomials over $\mathbb{F}_2$, whose number is given by the standard Möbius formula [5, Thm. 1.3.6]. The point at infinity adds one degree-one point, hence

$$a_1 = 3, \quad a_d = \frac{1}{d} \sum_{\substack{e|d \\ e \geq 2}} \mu_{\text{Mob}}(e) 2^{d/e} \quad (d \geq 2). \quad (3)$$

Lemma 1 (Degree support). *For every $d \geq 1$, $a_d > 0$.*

Proof. The degree-one claim follows from $0, 1, \infty$. If $d \geq 2$, then

$$\begin{aligned} da_d &= 2^d + \sum_{\substack{e|d \\ e \geq 2}} \mu_{\text{Mob}}(e) 2^{d/e} \\ &\geq 2^d - \sum_{\substack{e|d \\ e \geq 2}} 2^{d/e} \geq 2^d - \sum_{j=1}^{\lfloor d/2 \rfloor} 2^j \\ &= 2^d - 2^{\lfloor d/2 \rfloor + 1} + 2 > 0. \end{aligned}$$

The last expression equals 2 when $d = 2$ and is positive thereafter. $\square$

The exponential size of a_d is not used below. The assertion $a_d \geq 1$ for each degree is already enough to generate dense point spectrum and infinite multiplicity. Thus deleting all but one closed point in every degree would not repair the obstruction, although that deletion would define a different classical object.

4 Invariant measures and one Hilbert class

The canonical measure is counting measure on the discrete base times Lebesgue flow time, passed to the quotient. On the circle decomposition it is simply

$$\mu_1 = \sum_x du_x.$$

More generally, fix any family $w = (w_x)_x$ with $0 < w_x < \infty$, and put

$$\mu_w = \sum_x w_x du_x, \quad \mathcal{H}_w = L^2(M_F, \mu_w) = \bigoplus_x^2 L^2(C_x, w_x du_x). \quad (4)$$

Proposition 2 (Measure properties). *Every μ_w is invariant, sigma-finite, Radon, and has full support.*

Proof. Translation preserves du_x , hence every constant multiple $w_x du_x$. The component set is countable and each circle has finite weighted measure, proving sigma-finiteness. Strict positivity gives positive measure to every nonempty open subset of each component, so the measure has full support. A compact subset of a topological coproduct meets only finitely many open-and-closed components. Its measure is therefore a finite sum of finite circle measures, which proves the Radon property. $\square$

Theorem 3 (Positive-weight unitary equivalence). *Define*

$$W_w : \mathcal{H}_w \longrightarrow \mathcal{H}_1, \quad (W_w f)_x = \sqrt{w_x} f_x. \quad (5)$$

Then W_w is unitary and intertwines the Koopman groups. After the generators are defined in Section 5, it also maps their domains onto one another and satisfies $W_w A_w = A_1 W_w$.

Proof. The norm identity

$$\|W_w f\|_{\mathcal{H}_1}^2 = \sum_x \int_{C_x} w_x |f_x|^2 du_x = \|f\|_{\mathcal{H}_w}^2$$

holds, and componentwise multiplication by $w_x^{-1/2}$ is the inverse. Constant multiplication commutes with translation and differentiation. The same identity applied to f'_x proves the graph-norm and domain assertions. $\square$

Choosing positive weights with $\sum_x w_x L_x = 1$ makes μ_w a probability measure, and $w_x = L_x^{-1}$ gives normalized Haar measure on every component. Both are in the same unitary class. A zero component weight is excluded: it deletes closed-point data and is therefore a new candidate, not a reweighting.

5 Koopman group and self-adjoint generator

Freeze the inverse-flow pullback convention

$$(U_t^{(w)} f)_x(u) = f_x(u - t), \quad U_t^{(w)} = e^{-itA_w}. \quad (6)$$

The measure invariance in Proposition 2 gives unitarity. This is the usual Koopman construction for a measure-preserving flow [7].

Proposition 4 (Strong continuity). *The family $(U_t^{(w)})_{t \in \mathbb{R}}$ is a strongly continuous unitary group.*

Proof. Translation is strongly continuous on each circle L^2 -space. Given $f \in \mathcal{H}_w$ and $\varepsilon > 0$, choose a vector g , supported on finitely many components, with $\|f - g\| < \varepsilon$. Then

$$\|U_t^{(w)} f - f\| \leq 2\|f - g\| + \|U_t^{(w)} g - g\|.$$

The last term tends to zero in a finite direct sum. Since ε is arbitrary, the claim follows. $\square$

The formal derivative is insufficient until its domain and boundary conditions are fixed. Define

$$A_w = \bigoplus_x A_x, \quad A_x = -i \frac{d}{du}, \quad (7)$$

on

$$\mathcal{D}(A_w) = \left\{ f = (f_x) : f_x \in H_{\text{per}}^1(0, L_x) \text{ for every } x, \right. \\ \left. \sum_x w_x (\|f_x\|_{L^2(du_x)}^2 + \|f'_x\|_{L^2(du_x)}^2) < \infty \right\}. \quad (8)$$

The periodic condition means $f_x(0) = f_x(L_x)$ in the Sobolev trace sense.

Theorem 5 (Complete operator and Stone generator). *The operator A_w on (8) is self-adjoint and is the unique Stone generator of (6). Finite-component trigonometric polynomials form a core.*

Proof. Periodic Fourier transform gives the normalized component basis

$$e_{x,n}^{(w)}(u) = (w_x L_x)^{-1/2} e^{2\pi i n u / L_x}, \quad n \in \mathbb{Z}, \quad (9)$$

and

$$A_x e_{x,n}^{(w)} = \frac{2\pi n}{L_x} e_{x,n}^{(w)}. \quad (10)$$

Thus A_x is unitarily equivalent to multiplication by the real sequence $(2\pi n / L_x)_{n \in \mathbb{Z}}$ on its maximal domain, which is precisely periodic H^1 . It is self-adjoint. The orthogonal-sum theorem now gives a self-adjoint global operator on the maximal graph-summability domain (8) [8, Thm. 2.23].

Componentwise exponentiation produces translations. The sign is fixed by

$$\left. \frac{d}{dt} U_t^{(w)} f \right|_{t=0} = -f' = -i \left(-i \frac{d}{du} \right) f.$$

Stone's theorem gives uniqueness [6, 8]. Finite component truncation followed by finite Fourier truncation converges in the global graph norm, proving the core statement. $\square$

Theorem 5 establishes the two positive limited-route facts: B1_COMPLETE_OPERATOR_DEFINIT ION and B2_SELF_ADJOINT.

6 Exact point spectrum and degeneracy

Theorem 6 (Point spectrum). *For every positive weight family,*

$$\sigma_p(A_w) = \frac{2\pi}{\log 2} \mathbb{Q}. \quad (11)$$

Proof. A degree- d component has, by (10), the frequency lattice

$$\frac{2\pi}{d \log 2} \mathbb{Z}.$$

Lemma 1 supplies at least one component for every positive d , and the union of these lattices is $(2\pi / \log 2) \mathbb{Q}$.

Conversely, suppose $A_w f = \lambda f$ with $f \neq 0$. At least one component f_x is nonzero, and it satisfies $A_x f_x = \lambda f_x$. The component Fourier diagonalization forces $\lambda = 2\pi n / (d_x \log 2)$ for some $n \in \mathbb{Z}$. No other eigenvalues occur. $\square$

Theorem 7 (Countably infinite multiplicity). *Every $\lambda \in \sigma_p(A_w)$, including zero, has countably infinite multiplicity.*

Proof. Write $\lambda = (2\pi / \log 2)(a/b)$, with a/b in lowest terms and $b \geq 1$. For every $k \geq 1$, choose a closed point x_k of degree kb , possible by Lemma 1, and take Fourier mode ka . Then

$$A_w e_{x_k, ka}^{(w)} = \frac{2\pi}{\log 2} \frac{a}{b} e_{x_k, ka}^{(w)}. \quad (12)$$

The vectors have disjoint component supports and are orthonormal. For $a = 0$, the same construction is the constant mode on one component in each degree. Multiplicity is at most countable because the complete Fourier basis is countable. $\square$

The mechanism is not the large value of a_d . One point in every degree already provides the sequence (kb, ka) . Nor is it restricted to invariant functions: when $a \neq 0$, every witness in (12) is a nonconstant Fourier mode.

7 Full spectrum, essential spectrum, and terminology

Theorem 8 (Global spectral classification). *For every positive component-weight family,*

$$\boxed{\sigma(A_w) = \sigma_{\text{ess}}(A_w) = \mathbb{R}, \quad \sigma_{\text{disc}}(A_w) = \emptyset.} \quad (13)$$

Proof. The spectrum of a countable orthogonal sum is the closure of the union of the component spectra [8, Thm. 2.23]. Theorem 6 and density of $\mathbb{Q}$ give

$$\sigma(A_w) = \overline{(2\pi/\log 2)\mathbb{Q}} = \mathbb{R}.$$

Every neighborhood of a rational spectral point contains its infinite-dimensional eigenspace. If λ is irrational, choose distinct rational frequencies $\lambda_j \rightarrow \lambda$, and choose their normalized eigenvectors on mutually distinct components. Then

$$\|(A_w - \lambda)e_j\| = |\lambda_j - \lambda| \rightarrow 0, \quad e_j \rightharpoonup 0.$$

This is a singular Weyl sequence. Hence every real point is essential, and no isolated finite-multiplicity eigenvalue remains in the discrete spectrum [8, Secs. 6.2–6.4]. $\square$

Remark 9 (Two meanings that must not be conflated). The set $\{e_{x,n}^{(w)}\}_{x,n}$ is a complete orthonormal eigenbasis, so every vector spectral measure is atomic, or pure point. At the same time, the operator-theoretic set decomposition is

$$\sigma_p(A_w) = \frac{2\pi}{\log 2}\mathbb{Q}, \quad \sigma_c(A_w) = \mathbb{R} \setminus \frac{2\pi}{\log 2}\mathbb{Q}, \quad \sigma_r(A_w) = \emptyset. \quad (14)$$

For an irrational λ , $A_w - \lambda$ is injective with dense but nonsurjective range. Such points are continuous-spectrum accumulation points as a subset of the operator spectrum. They do not create a continuous component in any vector spectral measure. A complete pure-point basis is therefore compatible with dense spectrum and empty compact-resolvent discrete spectrum.

8 Compactness, counting, and heat obstructions

The global spectral theorem gives the exact Route-B obstruction.

Proposition 10 (Noncompact resolvent). *For every $z \in \mathbb{C} \setminus \mathbb{R}$, $(A_w - z)^{-1}$ is not compact. The failure remains on the orthogonal complement of $\ker A_w$.*

Proof. On every normalized zero mode,

$$(A_w - z)^{-1}e_{x,0}^{(w)} = -z^{-1}e_{x,0}^{(w)}.$$

The resolvent sends an infinite orthonormal sequence to a fixed nonzero multiple of itself, whose image has no norm-convergent subsequence. After deleting the kernel, choose any nonzero rational eigenvalue λ . Theorem 7 supplies an infinite orthonormal set on which the reduced resolvent is multiplication by $(\lambda - z)^{-1}$. $\square$

Table 2: Exact theorem ledger for the frozen generator.

Property	Result	Status
Point spectrum	$(2\pi/\log 2)\mathbb{Q}$	PROVED
Point multiplicity	countably infinite at every point eigenvalue	PROVED
Full and essential spectrum	both equal $\mathbb{R}$	PROVED
Discrete spectrum	empty	PROVED
Vector spectral measures	pure point; complete eigenbasis	PROVED
Irrational operator-spectrum points	continuous spectrum, no eigenvectors	PROVED
Resolvent	noncompact, also after kernel deletion	PROVED
Positive-width interval projections	infinite rank	PROVED
Heat and resolvent powers	not trace class	PROVED
Ordinary generator determinant	standard required inputs unavailable	PROVED boundary

Proposition 11 (No locally finite eigenvalue count). *If $I \subset \mathbb{R}$ is an interval of positive width, then*

$$\text{rank } \mathbf{1}_I(A_w) = \infty. \quad (15)$$

Consequently

$$N(E) := \dim \text{Ran } \mathbf{1}_{[-E, E]}(A_w) = \infty \quad (E \geq 0). \quad (16)$$

Proof. Every positive-width interval contains a member of $(2\pi/\log 2)\mathbb{Q}$, and its eigenspace is infinite-dimensional. For $E = 0$, the centered projection is the infinite-dimensional kernel. The word “interval” cannot be replaced by an arbitrary nonempty Borel set: a singleton irrational set has spectral-projection rank zero. $\square$

Proposition 12 (No trace-class heat). *For every $t > 0$, neither $e^{-tA_w^2}$ nor $e^{-t|A_w|}$ is trace class. The same holds after zero-mode deletion. Moreover, $(1 + A_w^2)^{-s/2}$ is not trace class for any $s > 0$.*

Proof. Both heat functions equal one on the infinite zero eigenspace. After its removal, fix a nonzero rational eigenvalue λ . Its infinite eigenspace receives the same positive multiplier $e^{-t\lambda^2}$, or $e^{-t|\lambda|}$. The resolvent power similarly repeats the positive number $(1 + \lambda^2)^{-s/2}$ infinitely. Each operator is therefore noncompact and not trace class. $\square$

The symbol e^{-tA_w} is not used as a heat operator: A_w is unbounded below. The standard semigroup choices are functions of A_w^2 or $|A_w|$, and those already fail.

Bornemann’s account of the ordinary operator determinant defines $\det(I + zK)$ canonically for trace-class K and relates the usual product, Plemelj, and exterior-power descriptions [1, Secs. 2–3]. The frozen resolvent, heat functions, and Koopman group do not supply such a trace-class input. A compact-resolvent spectral-zeta determinant is also unavailable because every positive-width window has infinite rank. This is a boundary for the standard constructions. It does not assert that no relative, compressed, or renormalized determinant can ever be defined; each would require additional choices and would constitute a new candidate.

9 Why the orbit zeta is not this spectral determinant

Three constructions share arithmetic input but not an operator. The orbit ledger is

$$Z_{\text{orb}}(s) = \prod_x (1 - e^{-sd_x \log 2})^{-1}. \quad (17)$$

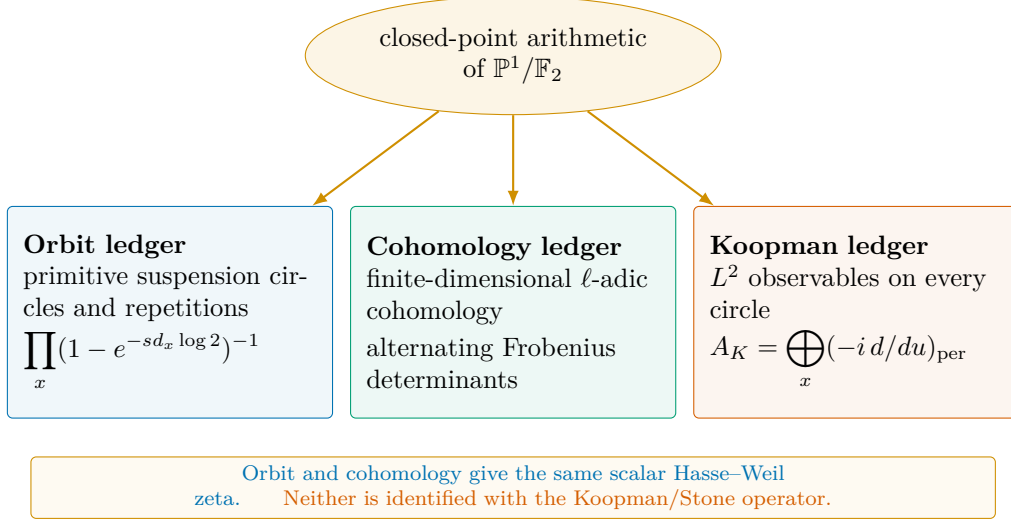


Figure 2: The shared closed-point source explains the scalar agreement of the orbit and cohomological Hasse–Weil constructions. It does not identify either operator with the Koopman/Stone generator.

It counts one primitive suspension circle per closed point and all positive repetitions. The Koopman ledger is

$$\mathcal{H}_w = \bigoplus_x^2 L^2(C_x, w_x du_x), \quad A_w = \bigoplus_x (-i d/du)_{\text{per}}, \quad (18)$$

which counts positive, negative, and zero Fourier modes on every circle. Nothing in (17) identifies its repetition coefficients with a trace of (18).

The third ledger is Deligne’s alternating cohomological expression

$$Z(X, t) = \prod_i \det \left(I - t \Phi_{\text{coh}} \mid H_c^i(X_{\overline{\mathbb{F}}_2}, \mathbb{Q}_\ell) \right)^{(-1)^{i+1}}, \quad (19)$$

where Φ_{coh} is the appropriate Frobenius action on compactly supported ℓ -adic cohomology [2, Eq. (1.5.4)]. These spaces are finite-dimensional in the present case and have an alternating cohomological grading. They are not the circle-observable Hilbert space, and Φ_{coh} is not the periodic derivative.

Proposition 13 (Same-object non-identification). *The frozen data supply no unitary conjugacy, trace theorem, or determinant identity between A_w and the cohomological Frobenius action. Therefore the orbit Hasse–Weil zeta cannot be presented as an ordinary Fredholm or compact-resolvent spectral-zeta determinant of A_w .*

Proof. The spaces, actions, and spectral types in (18) and (19) are explicitly different. No bridge morphism is part of the candidate. Independently, Propositions 10–12 show that the standard trace-class or compact-resolvent inputs fail on the Koopman side. A scalar identity derived from common closed-point counts cannot supply the missing operator map [10]. $\square$

Table 3: Deterministic control suite. Finite regressions check the implementation; the infinite claims are proved in the preceding sections.

Control	Result	Evidentiary boundary
Unit tests	8/8 pass	code and artifact consistency
Closed-point counts	positive through degree 24	regression for Lemma 1
Fixed-point reconstruction	all rows match	primitive-degree ledger check
Frequency witnesses	six signed rationals, 12 witnesses each	finite check of (kb, ka) construction
Kernel-deletion witness	survives	nonzero multiplicity control
Weight intertwiner	norm error zero; max action error 2.29×10^{-16}	finite floating-point regression
Manifest exclusions	no zeros; no fitted parameters; B4/B5 not invoked	scope lock

10 Koopman representation is not a physical quantization

The candidate is a natural unitary representation of classical transport. It maps an observable f to its pullback along the flow and converts time translation into a self-adjoint Stone generator. This justifies the formal enum `A4_UNITARY_OR_SCATTERING_CANDIDATE`.

Physical or geometric quantization requires further frozen structure. A standard geometric-quantization package includes symplectic and prequantum data together with a polarization or an alternative rule that fixes the observable-to-operator map [4]. The present disconnected circle suspension supplies none of those data, no Planck normalization, and no physical observable assignment beyond Koopman transport. Accordingly, the paper does not award `A4_NATURAL_QUANTIZATION` or `A4_ROUTE_B_READY`. This is a statement about the frozen candidate, not an impossibility theorem for every enrichment or quantization scheme.

11 Controls, limitations, and route verdict

11.1 Deterministic controls

The companion standard-library Python program regenerates three finite control tables and a manifest. It checks Möbius degree counts through degree 24, reconstructs fixed-point counts, verifies finite prefixes of the degree- kb , mode- ka witnesses, and tests the weight intertwiner on a finite Fourier vector. The control suite ran eight tests, all passing.

The generated manifest is `results/koopman_spectral_manifest.json`. It records `target_zero_data_used: false`, an empty `fitted_parameters` array, and `b4_b5_invoked: false`. The run is deterministic, has no random seed, makes no network request, and reads no rational-prime or target-zero table. The command is

```
./experiments/reproduce.sh
```

from the Paper 5 directory.

Table 4: Route A/A4 and limited Route B/B1–B3 assessment.

Layer	Formal enum	Result and reason
A4	A4_UNITARY_OR_SCATTERING_CANDIDATE	PROVED: canonical same-clock Koopman lift
B1	B1_COMPLETE_OPERATOR_DEFINITION	PROVED: space, measure, action, domain, boundary conditions
B2	B2_SELF_ADJOINT	PROVED: Fourier and orthogonal-sum proof
B3	B3_FAIL	PROVED: dense, infinitely degenerate essential spectrum
B4	outside limited audit	NOT INVOKED; no verdict serialized
B5	outside limited audit	NOT INVOKED; no verdict serialized
Overall	ROUTE_B_REJECTED	Gate C; hilbert_polya_claim_allowed: false

11.2 Limitations

The base topology is an imposed discrete topology inherited from the classical positive control. The resulting space is a countable topological coproduct of neutral periodic circles, with no coupling or transverse hyperbolicity. The Koopman conclusion therefore applies to this precise candidate and its positive component-weight unitary class. It is not a no-go theorem for all Frobenius-inspired operators.

The paper analyzes the canonical uncoupled translation representation. Adding a potential, off-diagonal coupling, nonconstant density, new boundary condition, compression, finite-degree cutoff, or renormalization changes the operator and requires a new source and same-object audit. A finite cutoff does have compact resolvent, but no frozen cutoff-removal or determinant renormalization theorem is supplied.

The determinant conclusion is likewise scoped to ordinary Fredholm and compact-resolvent spectral-zeta mechanisms formed from the frozen generator. It does not exclude a separately defined relative determinant. Nor does it deny Deligne’s cohomological determinant, whose correctness is exactly why the operator ledger must be named precisely.

No Riemann zeros, zero statistics, fitted scale, spectral shift, phase, potential, or target-informed boundary condition enter the construction or the controls. The paper therefore makes no numerical or inferential claim about a Riemann-zero correspondence.

11.3 Formal route report

Table 4 serializes only the layers actually audited. B4 and B5 are listed as scope annotations, not as verdict enums.

The earlier Riemann-target Route-A rejection of the fixed finite-field clock is unchanged [9]. The successful construction of a self-adjoint Koopman operator does not repair that classical arithmetic support mismatch, and B3 independently rejects this operator as a locally finite spectral host.

12 Conclusion

The Frobenius suspension does possess a fixed and natural self-adjoint Koopman generator. Its exact spectrum is nevertheless dense and infinitely degenerate: the point spectrum is $(2\pi/\log 2)\mathbb{Q}$,

while the full and essential spectra are $\mathbb{R}$. This proves that the generator cannot serve as the required locally finite determinant host under the standard mechanisms.

Any subsequent trace interface must identify a different source-derived operator and provide a rigorous same-object morphism. Transplanting the Hasse–Weil orbit or cohomological ledger into A_K would erase precisely the distinction established here.

Reproducibility and declarations

Data and code availability. All source-lock notes, acquired-source hashes, proof records, deterministic Python code, tests, generated CSV controls, JSON manifest, TikZ sources, bibliography, and manuscript source are included in `papers/5-quantum-flow/`; acquired source artifacts and their SHA-256 hashes are listed in `notes/source_matrix.md`. No external dataset or target-zero dataset was created or analyzed.

Ethics statement. This theoretical and deterministic computational study involves no human participants, animals, intervention, personal data, or identifiable information. Institutional ethics review was not required.

CRedit authorship contribution statement. Liang Wang: Conceptualization, Methodology, Formal analysis, Investigation, Software, Validation, Data curation, Visualization, Writing—original draft, and Writing—review and editing.

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A Fourier and direct-sum details

On a circle of length L_x , the periodic Fourier transform sends f_x to coefficients $\widehat{f}_x(n)$. In the weighted component norm,

$$\|f_x\|^2 = w_x L_x \sum_{n \in \mathbb{Z}} |\widehat{f}_x(n)|^2, \quad \|f'_x\|^2 = w_x L_x \sum_{n \in \mathbb{Z}} \left| \frac{2\pi n}{L_x} \widehat{f}_x(n) \right|^2.$$

Thus the global domain can equivalently be written as

$$\left\{ (\widehat{f}_x(n)) : \sum_{x,n} w_x L_x \left(1 + \frac{4\pi^2 n^2}{L_x^2} \right) |\widehat{f}_x(n)|^2 < \infty \right\}.$$

On this maximal real-multiplication domain the operator is self-adjoint. This coefficient form also proves that the displayed Fourier vectors are complete and that all vector spectral measures are countable atomic sums.

B Singular Weyl sequences

For a rational spectral point λ , choose distinct components carrying the same eigenvalue and their normalized eigenvectors. They are orthonormal, hence weakly converge to zero, and $(A_w - \lambda)e_j = 0$.

For an irrational λ , select pairwise distinct rational numbers q_j such that $(2\pi/\log 2)q_j \rightarrow \lambda$. Each rational frequency has infinitely many component realizations, so the associated eigenvectors can be placed on distinct components. This yields an orthonormal sequence and

$$\|(A_w - \lambda)e_j\| = \left| \frac{2\pi}{\log 2}q_j - \lambda \right| \rightarrow 0.$$

The construction proves essentiality at every real point and, at irrational points, failure of surjectivity of $A_w - \lambda$. Self-adjointness gives $\overline{\text{Ran}(A_w - \lambda)} = (\ker(A_w - \lambda))^\perp = \mathcal{H}_w$, so the range is dense.

C Artifact and integrity map

Relative path	Role
<code>notes/research_protocol.md</code>	frozen question, exclusions, sources, and proof obligations
<code>notes/source_matrix.md</code>	bibliographic identities, hashes, locators, and claim boundaries
<code>notes/candidate_lock.md</code>	immutable classical and operator definition
<code>notes/proof_audit.md</code>	complete theorem proofs and adversarial controls
<code>notes/composition_blueprint.md</code>	manuscript claim and architecture boundary
<code>notes/manuscript_integrity_audit.md</code>	post-build citation, claim, and reproducibility audit
<code>code/koopman_spectral_controls.py</code>	exact degree, frequency, and weight controls
<code>code/test_koopman_spectral_controls.py</code>	eight deterministic unit tests
<code>experiments/reproduce.sh</code>	one-command test and artifact regeneration
<code>results/closed_point_degree_controls.csv</code>	finite Möbius and fixed-point ledger
<code>results/frequency_multiplicity_witnesses.csv</code>	finite signed rational-frequency witnesses
<code>results/weight_unitary_controls.csv</code>	finite norm and intertwining regression
<code>results/koopman_spectral_manifest.json</code>	exclusions, hashes, theorem ledger, and Route scope
<code>paper/figures/</code>	source-native TikZ diagrams

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